AP Physics 1

Name _____

Period _____ Date _____

Practice Test – Introductory Dynamics (Version 4)

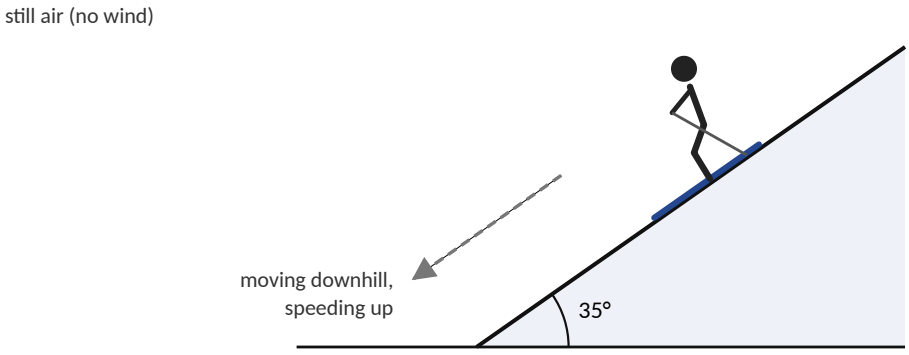
On any problems, you must show all work, include units, round properly, and box your final answer. Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

For all questions:

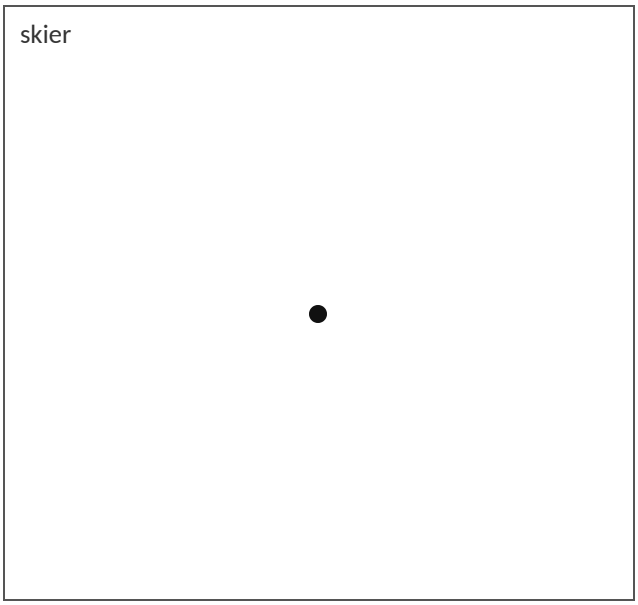
- Use g = 9.8 m/s² (9.8 N/kg). Air resistance is negligible unless the question says it is not.
- Free-body diagrams: draw the forces themselves, not components, each as an arrow starting on the dot.
- Round answers to the significant figures of the given values.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) A skier moves downhill on a snowy slope inclined at 35° to the horizontal. In the picture the skier moves down the slope to the left and is speeding up. Friction from the snow and drag from the air are **not** negligible.

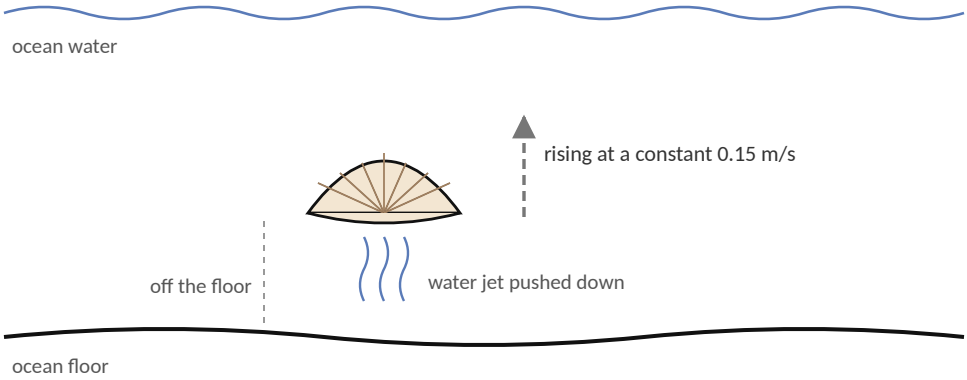


Assumptions for this practice problem: *the air is still (no wind), so the air moves past the skier opposite the skier's motion.*

Draw and label the free-body diagram of the skier. Draw the forces themselves, not components. Each force is one arrow that starts on the dot and points in the direction the force acts. No calculations.

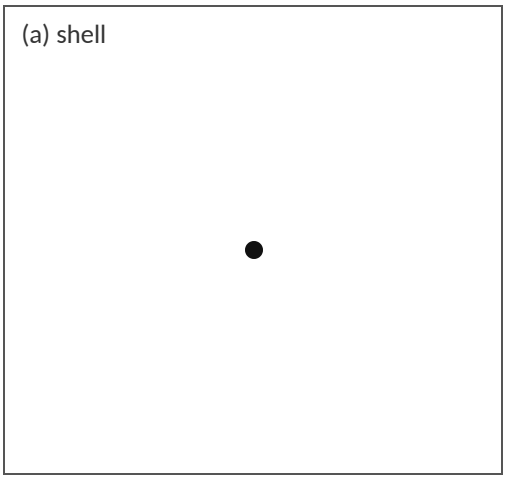


2. (25 points) A 0.400 kg shell has lifted off the ocean floor. It pushes water downward, so the water pushes it upward with a thrust of 2.50 N. After it leaves the floor, it rises at a constant speed of 0.15 m/s.



Assumptions for this practice problem: *water drag on the shell is negligible; $g = 9.8\ m/s^2$.*

(a) Draw the free-body diagram of the shell after it has left the floor.

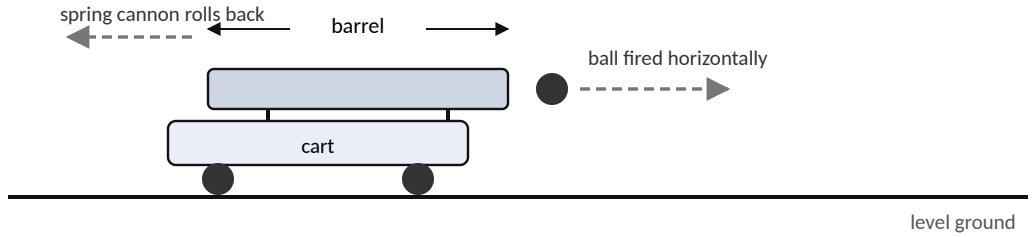


(b) Find the shell's acceleration.

(c) With up as positive, write Newton's second law for the shell and solve it for the buoyant force in letters.

(d) Calculate the magnitude of the buoyant force on the shell.

3. (25 points) A spring cannon fires a ball horizontally to the right. Find the spring cannon's acceleration while the ball is in the barrel.



ball mass	250. g
mass of the cannon and its cart	1.00 kg
barrel length	20.0 cm
exit speed, relative to the cannon	5.00 m/s

The exit speed is measured relative to the cannon, which is itself moving backward, so it is not the ball's speed relative to the ground.

Assumptions for this practice problem: *the spring cannon starts at rest on level ground and rolls freely (ignore friction and air resistance); the ball starts from rest; the ball's acceleration in the barrel is constant; the push between the ball and the spring cannon is the only horizontal interaction.*

(a) Draw the free-body diagram of the spring cannon while the ball is in the barrel.

(a) spring cannon

(b) Calculate the ball's acceleration relative to the cannon.

(c) Relative to the cannon, the ball's acceleration is the sum of the magnitudes of the ball's and the cannon's accelerations relative to the ground. Use Newton's second and third laws to find the ball's acceleration relative to the ground, and the force the cannon exerts on the ball.

(d) Calculate the cannon's acceleration, magnitude and direction.

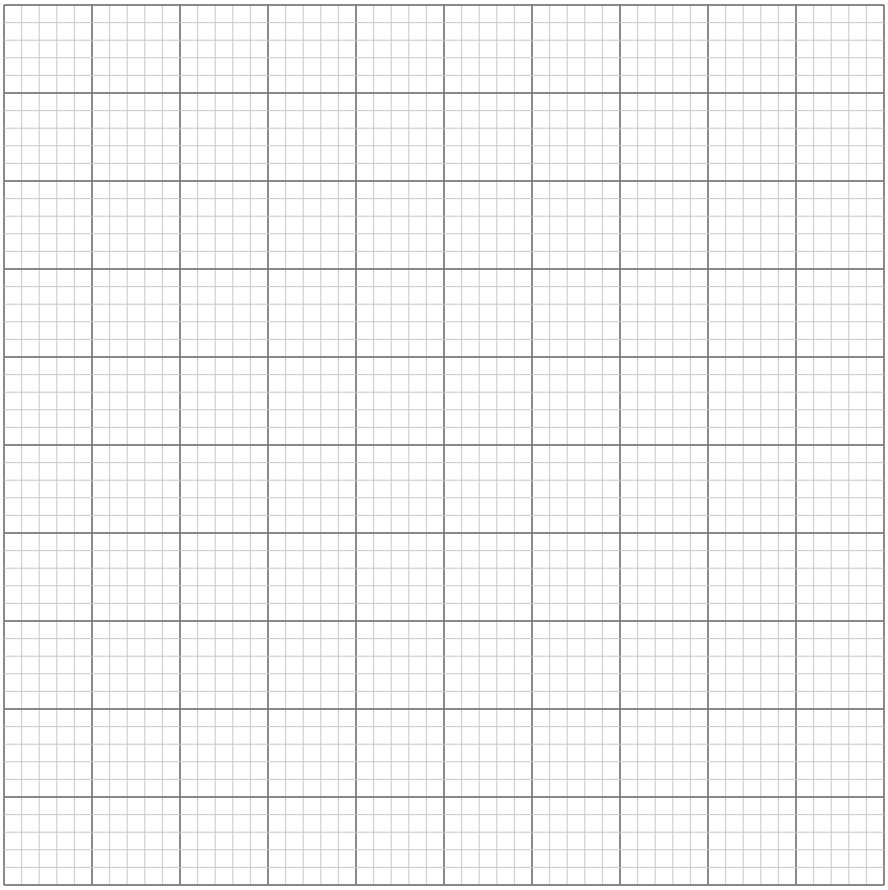
4. (25 points) Students measure the drag force F_D on a ski jumper in a wind tunnel at several air speeds v. The drag force is modeled by $F_D = Kv^2$, where K is a constant.

Speed v (m/s)	**Drag force F_D (N)**	
10.0	15.3	
14.0	30.5	
18.0	48.3	
22.0	68.9	
26.0	97.9	
30.0	132	

(a) Using $F_D = Kv^2$, state the quantity you will plot on the vertical axis and the quantity you will plot on the horizontal axis so the data form a straight line, each with its units. State what the slope equals.

(b) Use the blank column in the table to calculate the values you will plot on the horizontal axis.

(c) Plot the data on the grid. Label each axis with the quantity and units, choose a scale, and draw the best-fit line.



(d) Calculate the slope of your best-fit line using two well-separated points on the line. Then give K with its units.

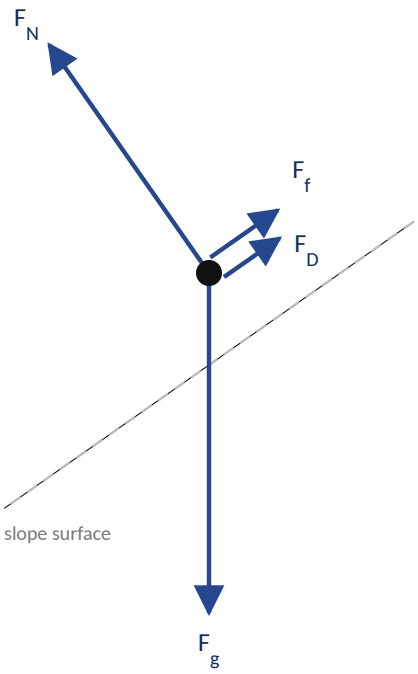
Answer Key – Practice Test, Version 4 – Introductory Dynamics

Worked step by step: diagram, symbolic equations and algebra first, then numbers with units. $g = 9.8\ m/s^2$ does not limit significant figures; answers match the significant figures of the givens.

Problems 2 and 3: diagram first, then a = ΣF/m (equation sheet), written a = F_{net}/m. Replace F_{net} with the actual forces and their signs, swap $F_g = mg$, solve in letters, then plug in numbers with units.
Kinematics: #1 $v = v_0 + at$ · #2 $\Delta x = v_0t + \frac{1}{2}at^2$ · #3 $v^2 = v_0^2 + 2a\Delta x$. Pick the one that leaves out the variable you don't know and don't need.

1. Skier free-body diagram (25 points)

skier, moving downhill to the left



Step 1 **List every force on the skier: exactly four.**

Earth pulls the skier: F_g. The snow pushes on the skier two ways: F_N (perpendicular to the surface) and kinetic friction F_f (along the surface). The air pushes on the skier: drag F_D.

Step 2 **Directions.**

F_g: straight down, vertical, not perpendicular to the slope.

F_N: perpendicular to the slope, pointing out of the snow, so it tilts 35° from vertical toward the left.

F_f: along the slope, uphill (up and to the right), opposite the sliding.

F_D: opposite the skier's motion through the air. The air is still and the skier moves downhill, so drag also points uphill, parallel to F_f.

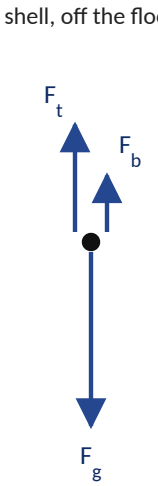
Step 3 **Check the diagram.**

Four arrows, each starting on the dot, each labeled. F_f and F_D are two separate arrows, side by side, both uphill. Speeding up, slowing down or moving at a constant speed changes the arrow lengths, not their directions.

Lengths in this key: speeding up downhill, so the two uphill arrows together are shorter than the part of gravity that pulls along the slope, and FN is a little shorter than Fg. Lengths are not graded.

Watch out: do not add the components of F_g (along and perpendicular to the slope) as extra arrows; they are not new forces. Drag is the air resistance, not a fifth force, and there is no forward "motion" force.

2. Shell: thrust and buoyancy (25 points)



Step 1 **(a) Forces on the shell after it leaves the floor.**

F_g down (Earth). Thrust F_t up (the water it pushes down pushes back up on it). Buoyancy F_b up (the surrounding water).

No normal force: the shell is no longer touching the floor. Constant velocity means a = 0, so $F_t + F_b$ together equal F_g.

Step 2 **(b) Acceleration from the motion data.**

Constant speed in a straight line means the velocity is not changing: **a = 0**

Here a = 0 comes from the stated constant speed, not from the fact that it moves upward.

Step 3 **(c) Newton's second law, up is +, letters first.**

$$a = \frac{F_{net}}{m} \rightarrow a = \frac{F_b + F_t - F_g}{m} \rightarrow F_b + F_t - mg = ma$$

$$\boxed{F_b = m(g + a) - F_t}$$

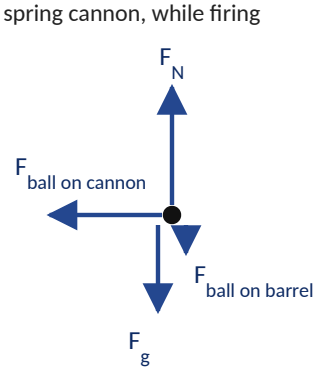
Step 4 **(d) Now the numbers, units in every parenthesis.**

$F_b = (0.400\ kg)(9.8\ m/s^2 + (0\ m/s^2)) - 2.50\ N = (0.400\ kg)(9.8\ m/s^2) - 2.50\ N = 3.920\ N - 2.50\ N = 1.420\ N$

$\boxed{F_b = 1.42\ N}$ (the subtraction leaves the hundredths place; g does not limit the significant figures).

Watch out: a = 0 here only because the speed is constant, not because the shell moves up. Dropping the thrust gives the wrong buoyant force.

3. Spring cannon recoil (25 points)



Step 1 **(a) Forces on the cannon while the ball is in the barrel.**

Vertical: the ground pushes up (F_N); Earth pulls down (F_g); the ball presses down on the barrel ($F_{ball\ on\ barrel}$). Nothing accelerates vertically, so $F_N = F_g + F_{ball\ on\ barrel}$.

Horizontal: the only force is the ball pushing back on the cannon, $F_{ball\ on\ cannon}$, to the left. The vertical forces do not enter the horizontal net-force equation. Arrow lengths are not to scale; the ball's push down is tiny next to the cannon's weight.

Step 2 **Units and frames.**

Convert first: 250. g = 0.250 kg; 20.0 cm = 0.200 m.

Step 3 **(b) Kinematics for the ball.**

Relative to the cannon: $v_0 = 0$, $v_{rel} = 5.00$ m/s over the barrel length $L = 0.200$ m. Use #3 in the cannon's frame.

$$a_{rel} = \frac{v_{rel}^2}{2L} = \frac{(5.00\ \text{m/s})^2}{2(0.200\ \text{m})} = \frac{25.0\ \text{m}^2/\text{s}^2}{0.400\ \text{m}} = \boxed{62.5\ \text{m/s}^2}$$

Step 4 **(c) Second and third laws, letters first.**

Third law: the ball and the cannon push on each other with forces of equal size F. Second law on each: $F = m_{ball}a_{ball}$ and $F = M\,a_{cannon}$, so $m_{ball}a_{ball} = M\,a_{cannon}$.

They accelerate in opposite directions, so relative to the cannon: $a_{rel} = a_{ball} + a_{cannon} = a_{ball} + \frac{m_{ball}a_{ball}}{M} \rightarrow$

$$a_{ball} = a_{rel}\frac{M}{M + m_{ball}}$$

$$a_{ball} = (62.5\ \text{m/s}^2)\frac{1.00\ \text{kg}}{1.00\ \text{kg} + 0.250\ \text{kg}} = \boxed{50.0\ \text{m/s}^2}$$ relative to the ground

$$F = m_{ball}a_{ball} = (0.250\ \text{kg})(50.0\ \text{m/s}^2) = \boxed{12.5\ \text{N}}$$

Step 5 **(d) Second law on the cannon.**

$$a_{cannon} = \frac{F_{\text{ball on cannon}}}{M} = \frac{12.5\ \text{N}}{1.00\ \text{kg}} = \boxed{12.5\ \text{m/s}^2\text{, to the left (backward)}}$$

Check: $50.0\ \text{m/s}^2 + 12.5\ \text{m/s}^2 = 62.5\ \text{m/s}^2 = a_{rel}$. After 0.0800 s the ball moves at 4.00 m/s and the cannon at 1.00 m/s backward, a relative speed of 5.00 m/s.

Watch out: the 5.00 m/s exit speed is relative to the cannon. Using it with the barrel length as if both were ground values gives 62.5 m/s² for the ball, 25% too big here, because this cannon is light.

4. Drag linearization (25 points)

Step 1 **(a) Match $F_D = Kv^2$ to $y = mx + b$.**

$F_D = K \cdot (v^2) + 0$

vertical (y)	horizontal (x)	slope	intercept
F_D (N)	v^2 (m²/s²)	K	0

Vertical: F_D in N. Horizontal: v^2 in m²/s². Slope = K.

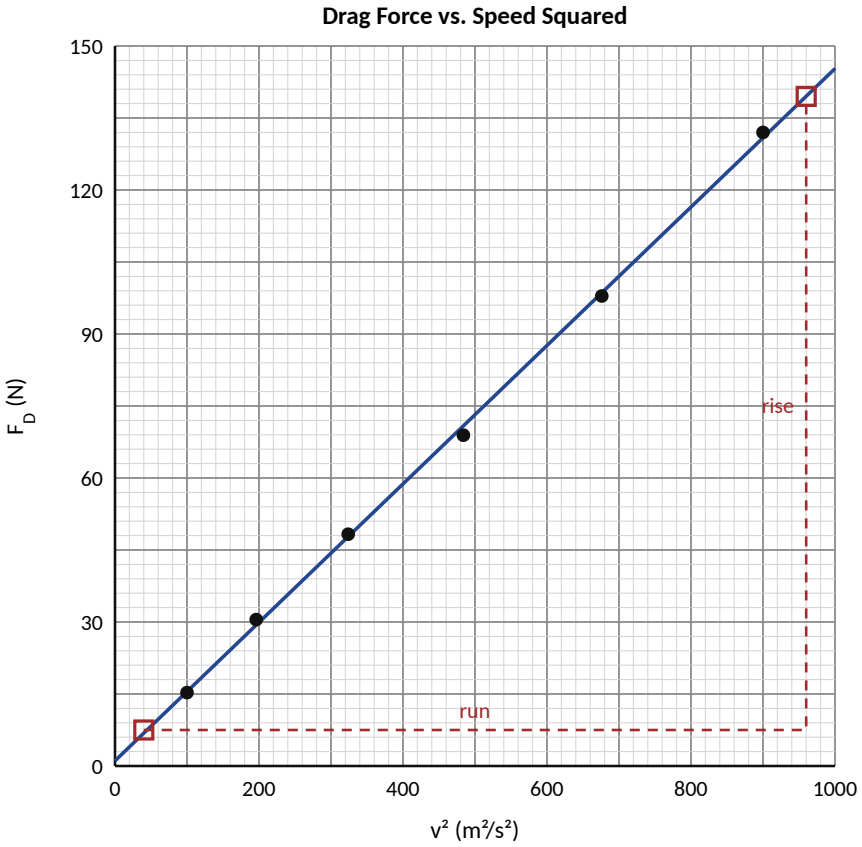
Watch out: v^2 is speed squared, in m²/s². It is not an acceleration (m/s²).

Step 2 **(b) Square each speed; the units square too.**

Example: $(10.0\ \text{m/s})^2 = 100\ \text{m}^2/\text{s}^2$

v (m/s)	F_D (N)	v^2 (m^2/s^2)
10.0	15.3	100
14.0	30.5	196
18.0	48.3	324
22.0	68.9	484
26.0	97.9	676
30.0	132	900

Step 3 **(c) Graph: labeled axes with units, a scale that uses more than half the grid, one straight line through the middle of the points.**



The best-fit line is not forced through any data point. The red squares are points read off the line, not data points.

Step 4 **(d) Slope from two well-separated points on the line, units carried.**

Points read off the line, away from data points: (40 m^2/s^2, 7.5 N) and (960 m^2/s^2, 139.5 N)

$$\text{slope} = \frac{139.5\ \text{N} - 7.5\ \text{N}}{960\ \text{m}^2/\text{s}^2 - 40\ \text{m}^2/\text{s}^2} = \frac{132.0\ \text{N}}{920\ \text{m}^2/\text{s}^2} = 0.1435\ \text{N}\cdot\text{s}^2/\text{m}^2$$

$K \approx 0.143\ \text{N}\cdot\text{s}^2/\text{m}^2\ (= \text{kg/m})$. Any value from 0.133 to 0.156 $\text{N}\cdot\text{s}^2/\text{m}^2$ from a reasonable line earns credit.

Units: $\text{N} \div (\text{m}^2/\text{s}^2) = \text{N}\cdot\text{s}^2/\text{m}^2$, and $\text{N} = \text{kg}\cdot\text{m}/\text{s}^2$, so $\text{N}\cdot\text{s}^2/\text{m}^2 = \text{kg/m}$.

1. Skier free-body diagram

Suggested partial credit	pts
F_g straight down (vertical), labeled	5
F_N perpendicular to the slope, out of the snow, labeled	5
F_f along the slope, uphill, labeled	5
F_D uphill (opposite the motion through still air), labeled, separate from F_f	5
Every arrow starts on the dot; no components and no extra forces	5
Total	**25**

2. Shell: thrust and buoyancy

Suggested partial credit	pts
(a) F_g down, thrust and buoyancy up, labeled	4
(a) No normal force	2
Acceleration: $a = 0$ from the constant speed	6
$F_b + F_t - mg = ma$ written with up as +	4
Solved in letters: $F_b = m(g + a) - F_t$	3
Substitution with units in every parenthesis (and grams to kg if needed)	3
$F_b = 1.42$ N	3
Total	**25**

3. Spring cannon recoil

Suggested partial credit	pts
(a) F_N up, F_g down (the ball pressing down may also be shown); ball-on-cannon force backward; no forward force	5
Unit conversions (g to kg, cm to m)	2
(b) $a_{rel} = v_{rel}^2/(2L) = 62.5$ m/s²	4
(c) $m_{ball}a_{ball} = M\,a_{cannon}$ from the second and third laws	4
(c) $a_{ball} = 50.0$ m/s² and $F = 12.5$ N	4
(d) $a_{cannon} = 12.5$ m/s², backward	6
Total	**25**

4. Drag linearization

Suggested partial credit	pts
(a) F_D (N) on the vertical axis, v^2 (m^2/s^2) on the horizontal axis	4
(a) Slope = K	2
(b) v^2 column correct, with units	4
(c) Axes labeled with quantity and units	2
(c) Scale uses more than half the grid	2
(c) Points plotted correctly	2
(c) One straight best-fit line	2
(d) Two well-separated points read off the line, not data points	3
(d) K from 0.133 to 0.156, with units $N \cdot s^2/m^2$ (or kg/m)	4
Total	**25**